

PROGRAM:

```
#include <stdio.h>

int main()
{
    int a, b, choice, res;

    printf("===== OPERATORS AND EXPRESSIONS =====\n");
    printf("Enter the first number: ");
    scanf("%d", &a);

    printf("Enter the second number: ");
    scanf("%d", &b);

    printf("\n----- MENU ----- \n");
    printf("1. Addition\n");
    printf("2. Subtraction\n");
    printf("3. Multiplication\n");
    printf("4. Division\n");
    printf("5. Modulus\n");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            res = a + b;
            printf("Result = %d", res);
            break;

        case 2:
            res = a - b;
```

```
printf("Result = %d", res);
```

```
break;
```

```
case 3:
```

```
    res = a * b;
```

```
    printf("Result = %d", res);
```

```
    break;
```

```
case 4:
```

```
    if(b != 0)
```

```
    {
```

```
        res = a / b;
```

```
        printf("Result = %d", res);
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Division by zero is not possible.");
```

```
    }
```

```
    break;
```

```
case 5:
```

```
    if(b != 0)
```

```
    {
```

```
        res = a % b;
```

```
        printf("Result = %d", res);
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Modulus by zero is not possible.");
    }
    break;
default:
    printf("Invalid choice.");
}
return 0;
}
```

OUTPUT

===== OPERATORS AND EXPRESSIONS =====

Enter the first number: 20

Enter the second number: 10

----- MENU -----

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus

Enter your choice: 1

Result = 30

PROGRAM:

```
#include <stdio.h>

int main()
{
    int a, b, res, choice;
    printf("==== BITWISE OPERATIONS ==== \n");
    printf("Enter the first number: ");
    scanf("%d", &a);
    printf("Enter the second number: ");
    scanf("%d", &b);
    printf("\n----- MENU ----- \n");
    printf("1. Bitwise AND (&) \n");
    printf("2. Bitwise OR (|) \n");
    printf("3. Bitwise XOR (^) \n");
    printf("4. Bitwise NOT (~) \n");
    printf("5. Left Shift (<<) \n");
    printf("6. Right Shift (>>) \n");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            res = a & b;
            printf("Bitwise AND Result = %d", res);
            break;
        case 2:
            res = a | b;
            printf("Bitwise OR Result = %d", res);
```

```
        break;
    case 3:
        res = a ^ b;
        printf("Bitwise XOR Result = %d", res);
        break;
    case 4:
        res = ~a;
        printf("Bitwise NOT Result = %d", res);
        break;
    case 5:
        res = a << b;
        printf("Left Shift Result = %d", res);
        break;
    case 6:
        res = a >> b;
        printf("Right Shift Result = %d", res);
        break;
    default:
        printf("Invalid choice.");
}
return 0;
}
```

OUTPUT

===== BITWISE OPERATIONS =====

Enter the first number: 12

Enter the second number: 5

----- MENU -----

1. Bitwise AND (&)
2. Bitwise OR (|)
3. Bitwise XOR (^)
4. Bitwise NOT (~)
5. Left Shift (<<)
6. Right Shift (>>)

Enter your choice: 1

Bitwise AND Result = 4